

Functional outcome swallowing scale*

Stage	Symptoms
0	Normal function and asymptomatic
1	Normal function with episodic or daily symptoms of dysphagia
2	Compensated abnormal function manifested by significant dietary modifications or prolonged mealtime (without weight loss or aspiration)
3	Decompensated abnormal function with weight loss of <10% of body weight over 6 months due to dysphagia; or daily cough, gagging or aspiration during meals
4	Severely decompensated abnormal function with weight loss of >10% of body weight over 6 months due to dysphagia; or severe aspiration with bronchopulmonary complications. Non oral feeding for most nutrition
5	Non oral feeding for all nutrition

*Salassa J.R. A functional outcome swallowing scale for staging oropharyngeal dysphagia. *Digestive Diseases* 1999; 17:230-234.

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2391 Intensity-Modulated Radiation Therapy (IMRT) in the Treatment of Cancer of the Larynx/Hypopharynx

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Purpose/Objective(s): Retrospective review of a single institution's IMRT experience in treating laryngeal/hypopharyngeal carcinoma.

Materials/Methods: Between 1/01 and 6/05, 38 patients (28 males, 10 females) with laryngeal/hypopharyngeal carcinoma underwent IMRT at our institution. Median age was 60 (range 36–78). There were 25 laryngeal, 13 hypopharyngeal carcinomas. The disease was Stage II in 2 (5%), Stage III in 11 (29%), and Stage IV in 25 (66%). Ten patients had T4a disease and one T4b. Concurrent chemotherapy was given to 34 patients (89%) and consisted of 2–3 planned cycles every 3 weeks of cis-platinum (20 pts), 2–3 planned cycles every 3 weeks of carboplatin/5-fluorouracil (11 pts), or weekly carboplatin/taxol (3 pts). All patients received definitive IMRT except for two who had post-operative IMRT (1 larynx, 1 hypopharynx). Median RT dose was 70 Gy. The majority of the patients were treated with an IMRT dose painting technique, and the prescription was defined as follows: 70 Gy was delivered to PTV70 (gross tumor volume + 5–10mm margin); 59.4 Gy to PTV59.4 (high risk subclinical region + 5–10mm margin); and 54 Gy to PTV54 (lower risk subclinical region + 5–10mm margin). Dose painting was not used in 5 patients treated with an earlier concomitant boost technique and in 2 patients receiving post-operative IMRT. Prophylactic feeding tubes (PEG) were placed in 33 patients. Acute toxicity and late xerostomia were scored using the RTOG radiation morbidity scale. The 2-year estimates for local progression-free rate (LPF), regional progression-free rate (RPF), locoregional progression-free (LRPF), distant metastases-free rate (DMF), and overall survival (OS) were calculated using the Kaplan-Meier method.

Results: With a median follow-up of living patients of 20 months (range 12 - 52), 2-year estimates of LPF, RPF, LRPF, DMF, and OS rates were 77%, 89%, 76%, 90%, 78%, respectively. All patients with local failure of larynx carcinoma had persistent disease at the end of treatment. Grade ≥ 2 mucositis/pharyngitis was experienced by 66% of the patients during treatment. Xerostomia continued to decrease over time from the end of RT and only 3 still complained of Grade 2 xerostomia at the time of this analysis. Fifty-four percent of patients with hypopharynx and 16% with larynx cancer were PEG dependent at the time of this analysis due to pharyngoesophageal stricture. One patient developed significant trismus and one developed laryngeal necrosis; both of these patients received concurrent chemoradiotherapy. Analysis of the dose-volume histograms showed that the average maximum, D05, mean, D95, and minimum doses delivered were 78 Gy, 75 Gy, 73 Gy, 69 Gy, and 56 Gy to PTV70; 76 Gy, 70 Gy, 64 Gy, 58 Gy, 34 Gy to PTV59.4; and 76 Gy, 70 Gy, 64 Gy, 58 Gy, and 34 Gy to PTV54.

Conclusions: To our knowledge, this is the first clinical report focused on the use of IMRT for laryngeal/hypopharyngeal carcinomas. These preliminary results show that IMRT achieved encouraging locoregional control in patients with laryngeal/hypopharyngeal carcinoma. Xerostomia improved over time. Pharyngoesophageal stricture with PEG dependency remains a problem in patients with hypopharyngeal carcinoma. Strategies using the ability of IMRT to limit the dose delivered to the esophagus and inferior constrictor musculature may be useful to further minimize this late complication.

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2392 Carbon Ion Radiotherapy for Uveal Melanoma

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Purpose/Objective(s): The aim of this study was to evaluate the efficacy and feasibility of carbon ion radiotherapy (C-ion RT) as a conservation method of eyeball for Japanese patients with locally advanced uveal melanoma.

Materials/Methods: Between April 2001 and July 2005, 57 patients received C-ion RT; 42 patients were enrolled in the dose escalation study (60, 70, 77 and 85 GyE in 5 fractions), thereafter 15 patients were treated with C-ion RT using a fixed dose of 70 GyE in 5 fractions as a treatment of a highly advanced medical technology from April 2004. Patients were eligible if retention of their visual acuity in patients was unexpected with the proton therapy due to the tumor size or tumor location,